

## EE3621: DIGITAL SIGNAL PROCESSING | III B.TECH EEE

### § Faculty Reference Document — Textbook Replacement

## PREFACE FOR FACULTY

In real-time DSP, input sequences  $x[n]$  (speech, audio, radar returns) are effectively infinite in length, while FIR filter impulse responses  $h[n]$  have finite length  $M$ . Computing the entire convolution at once is impossible due to memory limits and unacceptable latency. The **Overlap-Add (OLA)** method segments the input into non-overlapping blocks of length  $L$  and performs block convolution via FFT/IFFT.

### Pedagogical Strategy:

1. Formulate input signal decomposition into disjoint blocks:  $x[n] = \sum_{m=0}^{\infty} x_m[n - mL]$ .
2. Establish block zero-padding requirement:  $N \geq L + M - 1$ .
3. Compute frequency-domain circular convolution for each block:  $y_m[n] = \text{IDFT}_N\{\text{DFT}_N\{x_m\} \cdot \text{DFT}_N\{h\}\}$ .
4. Synthesize the total output by adding the overlapping  $M - 1$  tails between consecutive blocks.
5. Derive the optimal block length  $L_{\text{opt}}$  that minimizes total computational cost per output sample.

## 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Partition** long streaming sequences into blocks suitable for Overlap-Add filtering.
2. **Execute** complete numerical block filtering using the Overlap-Add method.
3. **Determine** the required FFT size  $N$  to prevent aliasing within blocks.
4. **Calculate** overall throughput and latency for real-time DSP implementations.

## 2. MATHEMATICAL FOUNDATIONS

### § 2.1 Overlap-Add Block Decomposition

Let  $x[n]$  be an infinite input sequence and  $h[n]$  an FIR filter of length  $M$ . Segment  $x[n]$  into non-overlapping blocks of length  $L$ :

$$x_m[n] = \begin{cases} x[n + mL], & 0 \leq n \leq L - 1 \\ 0, & \text{otherwise} \end{cases}$$

Total input:  $x[n] = \sum_{m=0}^{\infty} x_m[n - mL]$ .

### § 2.2 Linear Convolution of Sub-Blocks

By Linearity of LTI systems:

$$y[n] = x[n] * h[n] = \left( \sum_{m=0}^{\infty} x_m[n - mL] \right) * h[n] = \sum_{m=0}^{\infty} (x_m[n - mL] * h[n]) = \sum_{m=0}^{\infty} y_m[n - mL]$$

Where each block convolution  $y_m[n] = x_m[n] * h[n]$  has length:

$$N = L + M - 1$$

## § 2.3 FFT Implementation & Overlap-Addition

1. Choose FFT size  $N = 2^k \geq L + M - 1$ .
2. Zero-pad  $x_m[n]$  with  $N - L$  zeros to length  $N$ .
3. Zero-pad  $h[n]$  with  $N - M$  zeros to length  $N$ .
4. Precompute  $H[k] = \text{DFT}_N\{h[n]\}$ .
5. For each block  $m$ :
  - $X_m[k] = \text{DFT}_N\{x_m[n]\}$
  - $Y_m[k] = X_m[k] \cdot H[k]$
  - $y_m[n] = \text{IDFT}_N\{Y_m[k]\}$
6. Reconstruct output  $y[n]$ : Overlap the last  $M - 1$  points of block  $m$  with the first  $M - 1$  points of block  $m + 1$  and add.

## 3. WORKED NUMERICAL EXAMPLES

### § Example 13.1: Complete Numerical Overlap-Add Filtering

**Problem:** Filter input  $x[n] = \{1, 2, -1, 2, 3, -2, 0, 1, 2, 1\}$  using FIR filter  $h[n] = \{1, 2, 1\}$  ( $M = 3$ ) using Overlap-Add with block length  $L = 4$ .

**Solution:**

- $L = 4, M = 3 \implies N = L + M - 1 = 4 + 3 - 1 = 6$ . (Pad to  $N = 6$ ).
- Partition input into blocks of  $L = 4$ :
  - $x_0[n] = \{1, 2, -1, 2, 0, 0\}$  (padded with 2 zeros)
  - $x_1[n] = \{3, -2, 0, 1, 0, 0\}$
  - $x_2[n] = \{2, 1, 0, 0, 0, 0\}$
- Filter padded:  $h[n] = \{1, 2, 1, 0, 0, 0\}$ .
- **Block Convolutions** ( $y_m[n] = x_m * h$ ):
  - $y_0[n] = \{1, 2, -1, 2\} * \{1, 2, 1\} = \{1, 4, 4, 2, 3, 2\}$
  - $y_1[n] = \{3, -2, 0, 1\} * \{1, 2, 1\} = \{3, 4, -1, -1, 2, 1\}$
  - $y_2[n] = \{2, 1, 0, 0\} * \{1, 2, 1\} = \{2, 5, 4, 1, 0, 0\}$
- **Overlap-Addition** ( $L = 4$  shift):

$$\begin{array}{rcccccccccccc}
 y_0 : & 1 & 4 & 4 & 2 & 3 & 2 & & & & & & \\
 y_1 : & & & & & 3 & 4 & -1 & -1 & 2 & 1 & & \\
 y_2 : & & & & & & & & & 2 & 5 & 4 & 1 \\
 \hline
 y[n] : & 1 & 4 & 4 & 2 & 6 & 6 & -1 & -1 & 4 & 6 & 4 & 1
 \end{array}$$

Result:

$$y[n] = \{1, 4, 4, 2, 6, 6, -1, -1, 4, 6, 4, 1\}$$

## 4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

### § Question 1 (15 Marks)

(a) Describe the Overlap-Add method for linear filtering of long data sequences. Explain why zero-padding is necessary. (7 Marks) (b) Use the Overlap-Add method to filter  $x[n] = \{1, 2, 0, -1, 3, 1, 2, -1\}$  with  $h[n] = \{1, 1, 1\}$  using block length  $L = 4$ . (8 Marks)

#### Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Mathematical formulation of block decomposition and overlap additions (4 Marks)
  - Explanation of  $N \geq L + M - 1$  to prevent circular aliasing (3 Marks)
- **Part (b):**
  - Parameter formulation:  $L = 4, M = 3 \implies N \geq L + M - 1 = 6$ .
  - Block setup:
    - $x_0[n] = \{1, 2, 0, -1, 0, 0\}$  (padded to  $N = 6$ )
    - $x_1[n] = \{3, 1, 2, -1, 0, 0\}$  (padded to  $N = 6$ ) (2 Marks)
  - Block convolutions ( $y_m[n] = x_m * h$ ):
    - $y_0[n] = \{1, 2, 0, -1\} * \{1, 1, 1\} = \{1, 3, 3, 1, -1, -1\}$
    - $y_1[n] = \{3, 1, 2, -1\} * \{1, 1, 1\} = \{3, 4, 6, 2, 1, -1\}$  (4 Marks)
  - Overlap-Addition assembly ( $L = 4$  shift):

$$\begin{array}{rcccccccc} y_0 : & 1 & 3 & 3 & 1 & -1 & -1 & & \\ y_1 : & & & & & 3 & 4 & 6 & 2 & 1 & -1 \\ \hline y[n] : & 1 & 3 & 3 & 1 & 2 & 3 & 6 & 2 & 1 & -1 \end{array}$$

Result:

$$y[n] = \{1, 3, 3, 1, 2, 3, 6, 2, 1, -1\}$$

(2 Marks)

## 5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

x = np.array([1, 2, 0, -1, 3, 1, 2, -1])
h = np.array([1, 1, 1])
y = np.convolve(x, h)
print("Direct Linear Convolution:", y)
```